
HALO REACH: Design and optimisation of the pusher propeller

UAS for Disaster Relief - AE3 2024



Student: Parasdeep Singh

CID: 02099678

Supervisor: Dr Kostas Steiros

Department: Department of Aeronautics

Course: MEng Aeronautical Engineering

Module: AERO60004 - Group Design Project

Date: 17/06/2024

Academic Year: 2023-2024

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.



Table 1. People involved in HALO REACH - GDP

Team code	Team	Names	CID
FATT01	Team Leads	Moreno, Patricia Rashid, Hasan H	01857449 02020313
FATT02	Aerodynamics	Alam, Shahriar Dirirsa, Naboth Lawlor, Max F Singh, Parasdeep Homa, Matthew M	02037395 02026211 02026199 02099678 02038403
FATT03	Structures	Leng, Jiaxun (Aaron) Wang, Zhiyan Liu, Stephen Chauhan, Shivani	02015060 01911209 02015258 02096006
FATT04	Systems and propulsion	Lin, Yongkai Chursuk, Tharm Wankhade, Divij P Aggarwal, Ishaan K Halim, George	02029838 02048996 02027906 02030969 02059571
FATT05	Flight dynamics	Baikady, Thilak S Scott-Brown, Ruairidh Allen, Harry Tang, Shun S Lee, Jeongheon (Edward)	02016794 01734080 02028116 02023280 01754474
FATT06	Add-ons	Rosenthal, Adam T Ramteke, Shreya Nakhla, Andrew	02027024 02079050 02044646



Table 2. Separation of Work within Aerodynamics team

Topic	People Involved	Person Presenting
Initial Concepts and Concept Selection	All	Parasdeep
Constraint Diagram	Max	Max
Wing Aerofoil Selection	Max	Max
Main Wing Design	Matthew, Max	Matthew
Analytical Drag Estimates	Max	Max
Winglet Design	Naboth	Naboth
Adkins and Leibeck Code	Parasdeep, Naboth	Parasdeep
Propellor Initial Thrust/Power analysis	Parasdeep	Parasdeep
Propeller Design	Parasdeep	Parasdeep
Propeller performance analysis	Parasdeep	Parasdeep
Rotor Initial Thrust/Power analysis	Shahriar	Shahriar
Rotor Design	Shahriar	Shahriar
Rotor Performance Analysis	Shahriar	Shahriar
CFD Setup and Validation for 2D Aerofoil	Matthew, Max	Max
CFD Setup and Validation For Stock Rotor	Matthew, Max	Matthew
Rotor CFD	Shahriar, Matthew	Shahriar
Propeller CFD	Matthew	Matthew
Full Wing CFD	Naboth	Naboth
Full Aircraft CFD	Max	Max
Blanketing Affects of Tailplane	Naboth	Naboth
Tailplane CFD	Naboth	Naboth

Abstract

This report describes the design process and considerations for a specialised propeller for a novel hybrid UAV system which will be used in disaster relief in response to the devastating earthquakes that struck parts of Syria and southern and central Turkey, with an devastating death toll of 60000. The UAV design aims to reduce these death tolls in future disasters. Blade element momentum theory with aerodynamic optimisation is utilised throughout the report to find the optimal design of propeller blade, which is then compared and contrasted against open-source softwares in order to validate the results. Optimisation required examining through multiple constraints and translating these constraints into blade parameters, and then further refining the design using MATLAB code implementation. The objective of this GDP is to produce a preliminary draft of this UAV for disaster relief and aid. This report should be read in conjunction with all the relevant teams and team members throughout the report in order to understand the clear goals of each sub-stage of the project.

Contents

Table of Contents	iv
List of Figures	vi
List of Tables	vii
Nomenclature	viii
1 Introduction	1
2 Literature Review	1
3 Preliminary Analysis	2
3.1 Constraints	2
3.2 Power Requirement for Acceleration	2
3.3 Initial Thrust Analysis	2
3.4 Selecting Design Point	3
4 Design Considerations of Parameters	3
4.1 Tip speeds and non-dimensional quantities	3
4.2 Tip Diameter	4
4.3 Number of blades	4
4.4 Hub Radius	5
4.5 Aspect Ratio	5
4.6 Summary of the parameters	5
5 Optimisation of Propeller Design	6
5.1 Theory of optimisation	6
5.2 Design methodology	7
5.3 Optimisation using Adkin's and Liebeck Method	7
5.4 Aerofoil Selection	8
5.5 Results	9
6 Performance Analysis	10
6.1 Optimal Design and comparison with XROTOR	10
6.2 Performance testing using Qblade	11
6.3 Results from CFD simulations	11
7 Conclusion and Discussion	12
7.1 Further improvements	12
References	13
Appendix	a
A MATLAB code	a
A.1 Adkin's and Liebeck's implementation	a
A.2 Performance code for fixed geometry input	f
B Parameter selection	i
B.1 Initial Reynolds number and Mach number distribution	i
B.2 RPM selection	i
C Aerofoil Data from airfoil tools	j

D	Final blade profile and data	m
E	XROTOR results for optimised geometry	n
F	Performance analysis	n
G	Other CFD results	o

List of Figures

4	Final Blade CAD	1
5	Power required against time taken to accelerate to cruise speed 41.1 m/s with an estimated drag model	2
6	Plot of thrust against airspeed showing propeller efficiency required to meet given performance parameter in terms of thrust	3
7	Plot of thrust against airspeed showing thrust required to meet given climb rate at different air speeds	3
8	Estimated Reynolds number and Mach number distribution along a single propeller blade	3
9	Effect of airspeed (a) and blade area ratio (b) on the size of a n-bladed ($n = 3, 4, 5$) propeller when replaced with 2-bladed propeller	5
10	Forces, angles and velocity acting on the propeller blade[3]	6
11	Overall design approach taken for optimum propeller design	7
12	Adkin and Liebeck's optimisation method for propeller design	8
13	Optimised Reynolds number and Mach number distribution along a single propeller blade	9
14	Optimised chord and twist distribution along a single propeller blade generated from both in-house MATLAB and Qblade, using Adkin and Liebeck's Method	10
15	Propeller performance coefficients against advance ratio, for loiter(a) and cruise speed(b)	11
16	CFD simulation results for thrust and torque generated at different RPMs in loiter, compared to BEMT predictions	12
17	Initial Reynolds number and Mach number distribution along a single propeller blade	i
18	Efficiency against speed-RPM using optimisation code	j
19	$\frac{L}{D}$ against Re for different aerofoils considered	j
20	Angle of attack at which the max L/D is achieved against Re for different aerofoils considered	k
21	C_L at which the maximum L/D is achieved against Re for different aerofoils considered	k
22	$C_{L_{max}}$ against Re for different aerofoils considered	l
23	Stall angle at which maximum $C_{L_{max}}$ is achieved against Re for different aerofoils considered	l
24	Finalised single blade for the propeller	m
25	Optimised local C_L and local angle of attack distribution along a single propeller blade	m
26	Finalised single blade for the propeller	n
27	CFD simulation results for thrust and torque generated at different RPMs in cruise, compared to BEMT predictions	o

List of Tables

1	People involved in HALO REACH - GDP	i
2	Separation of Work within Aerodynamics team	ii
3	Initial propeller tip diameter estimation	4
4	Target Design point	6
5	Comparison of aerofoils considered for the propeller blade cross section	9
6	Optimal design values comparison with XROTOR results	10

Nomenclature

$(L/D)_{max}$	Maximum lift-to-drag ratio
$(t/c)_{max}$	Maximum Thickness-to-chord ratio
a	axial interference(induction) factor
a'	rotational interference(induction) factor
M_{DD}	Drag-divergence Mach number
$(T/W)_{min}$	Minimum thrust to weight ratio
α, AoA	Angle of attack
$\mathcal{R}$	Aspect ratio
ϵ	drag-to-lift ratio
η_P	Propeller efficiency
κ_a	Korn's factor
$\Lambda_{0.5}$	Midchord sweep angle
ϕ	Flow angle
σ	Solidity
ξ	Non-dimensional blade radius
ζ	Displacement velocity factor
BEMT	Blade element momentum theory
c	Local chord length
CFD	Computational fluid dynamics
CG	centre of gravity
D	Blade tip diameter
F	Prandtl's tip loss factor
G	Circulation function
HALO REACH	Highly Aerodynamic Light Operation Remote Emergency Aid & Crisis Help
MTOW	Maximum Takeoff Weight
n	Revolutions per second
N_B	number of blades
R	Blade tip radius
R_{Hub}	Hub radius
Re	Reynolds number
RPM	Revolutions per minute
S	Blade surface area
UAS	Uncrewed Aerial System
V_{cruise}	Cruise velocity
V_{loiter}	Loiter velocity
VTOL	Vertical Take-Off and Landing
W	Local total velocity

1 Introduction

This report will talk the reader through the design, development and optimisation of the pusher propeller used in the UAV for disaster relief, by using aerodynamic considerations and optimisation methods. The aim is to find an optimum geometry of a propeller which works efficiently for different mission requirements, including climb, cruise and loiter. The pusher propeller is a critical part for the UAV performance, as the UAV is a hybrid configuration of an VTOL and conventional aircraft, that requires a thrust producing powerplant for fixed-wing forward flight. This report compares analytical BEMT with publicly available softwares (XROTOR and QBlade) and CFD results to validate the propeller design and performance. While optimising, a number of parameters, including blade diameter, number of blades, rotational speeds, hub diameter, flight speeds, etc, were considered, in order to obtain the optimum parameters to generate a thrust of 190 N for loiter speed of 27.6 m/s, which is chosen as the design point from all the critical parts of the flight, as the loiter takes up almost 87% of the whole mission endurance. The report also references to other teams and team members reports as the design is inter-linked and constrained from multiple design perspectives, which gives the reader a clear picture of overall thinking process.



Figure 4. Final Blade CAD

2 Literature Review

Initially, a turbojet was considered as using jet fuel is one of the requirements of the project, however, turbojet poses a big challenge on the weight restrictions for producing the required thrust for this type of UAV. Then, the propulsion team [1] managed to obtain a Wankel(Rotary hybrid) powerplant called RW1-79 [2], which is much lighter and provides a max power output of 18 kW(more than enough for this mission's requirements), hence a design for propeller was opted. A propeller is a lot more efficient means of converting engine power into propulsive power than jet engine for a UAV[3].

As the initial sizing and the concept was finalised, different configurations of the propeller were considered. Mainly, pushers and tractors were compared and contrasted. Similar configurations were considered in order to make better assumptions and judgements. This included SkyEye Sierra [4], an hybrid VTOL UAV, that provides very similar design requirements and uses a pusher. Another existing design considered was ThunderB-VTOL [5], which has been developed for long-endurance, while using a similar configuration. For our mission purposes, and considering a hybrid concept of VTOL UAV with fixed wing, that is capable of both VTOL and long endurance, pusher propeller was considered a better option over the tractor, due to few key reasons. The pusher provides an energised flow in front of the propeller, which further reduces the flow separation on the body at higher angle of attacks, whereas a tractor generates turbulent high speed wake over the fuselage which increases the overall drag[3]. For tractors, there could be structural vibrations at some specific conditions, which results in damaging the air frame. On the other hand, pusher, gives better stability as the normal force is aft of the aircraft CG. In conceptual design, it was also decided to have boom mounted twin-tail, and hence it makes the pusher ideal for placement. Some drawbacks of using pusher include ground clearance, which will be considered in the design. The blanketing effects of the fuselage may distort the flow, which might lead to unbalanced disc loading and mismatched structural stresses, which will be analysed using CFD simulations. There will also be increased broadband noise[3] due to the ingestion of fuselage wake, which can be considered in next iterations of the design.

3 Preliminary Analysis

After the conceptual design is finalised, a number of constraints needed to be analysed, which restricts and affects the propeller design. These constraints were further converted into design parameters and target specifications.

3.1 Constraints

The pusher propeller is subject to a number of design constraints, which were influenced by other teams. One of them is to keep the blade diameter below 1 m, due to ground clearance requirement prescribed in conceptual design from structures team[6], which also prevents potential propeller strike with the ground and the VTOL rotors. Another constraint is set out by both flight dynamics and systems team[1] to keep the total propeller mass below 0.7 kg due to stability considerations. The minimum propeller blade thickness is set to be at least 1 mm, prescribed by structures team[7] for better structural integrity and manufacturability, as composites will be used for propellers. Finally, the propeller should be able to generate required thrust for different phases of flight using only 6-8 kW from the engine, selected by the propulsion team[1] for better fuel efficiency.

3.2 Power Requirement for Acceleration

After the initial take-off phase, the UAV needs to accelerate to the cruise speed, which is 41.1 m/s, before it starts its climb phase. The power required for this phase is dependent on the acceleration rate, which directly affects the overall drag on the UAV. Fig.5 shows the effect of increasing the time taken to accelerate on the power required. From the engine power constraints, the design will be aimed to achieve this acceleration in 61 seconds, with a power required of 7 kW, which is calculated using an estimated overall drag model of the UAV from empirical models [8], which can be further refined in later iterations. This power requirement gives a target power required, which will be used to test the performance in different phases of mission.

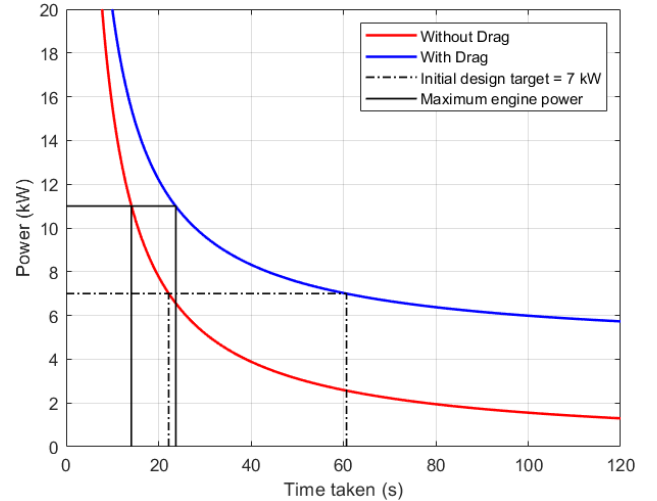


Figure 5. Power required against time taken to accelerate to cruise speed 41.1 m/s with an estimated drag model

3.3 Initial Thrust Analysis

Before initiating the propeller design, it is also important to define the required thrust for the UAV. By considering the same drag model as the power required for acceleration, and examining similar configurations [8], and the thrust analysis is performed to find desirable performance characteristics using an airspeed-thrust map shown in fig. 6. From the airspeed-thrust map, a desired η_P of 0.8 is chosen as a ballpark value, according to [9], to find the thrust required at different phases of flight, mainly at cruise and loiter, which is found to be 190 N (blue dot) and 270 N, respectively. The map also highlights the levelled flight thrust required which is 90 N in loiter, when thrust is equal to drag (red dot). From the initial constraint diagram [10], the $(T/W)_{min}$ at loiter speed of 27.6 m/s is found to be 0.117, which equates to 184 N of thrust, using maximum take off weight of 160 kg. The maximum static thrust, which is thrust generated at maximum power provided by engine, when the UAV is stationary, is considered in the initial thrust estimations, as it provides physical limit to work with. This is further corrected for blockage effects and presence of hub, and presented in the plot, using an empirical model [3].

Another airspeed-thrust map is produced for estimating the thrust required in climbing phase. Fig. 7 shows the change in thrust required for a given flight velocity at different climb rates. It can

be deduced that when climbing at the minimum allowable climb rate of 2.05 m/s given in the design brief, a thrust of 256 N will be required when we start climbing from cruise speed of 41.1 m/s.

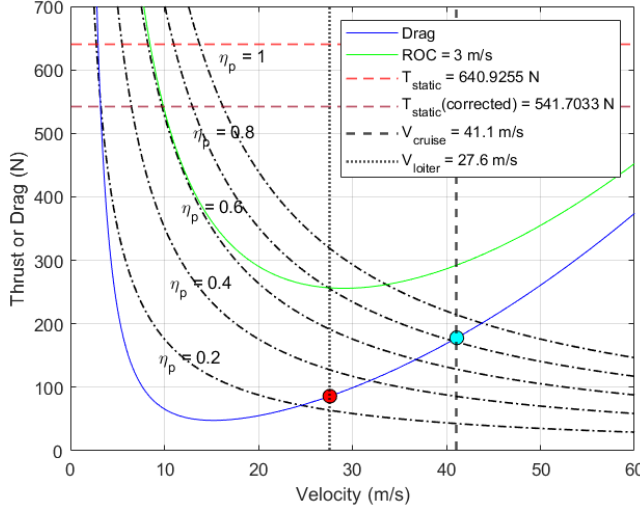


Figure 6. Plot of thrust against airspeed showing propeller efficiency required to meet given performance parameter in terms of thrust

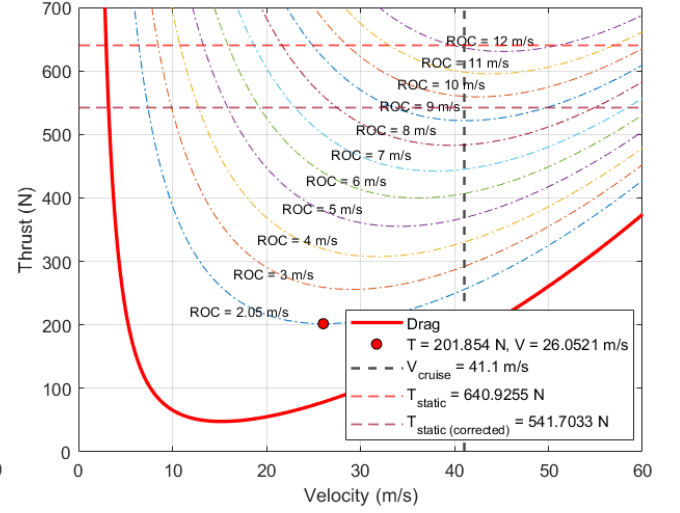


Figure 7. Plot of thrust against airspeed showing thrust required to meet given climb rate at different air speeds

3.4 Selecting Design Point

Considering all different phases of mission, and their corresponding thrust requirements, it is decided to select the loiter as the design point for the propeller design, since loiter is the most prolonged(endurance) phase, where it is required to loiter for more than 10 hours at 500 m. This demands the propeller to be most efficient in this phase and then analysis of the performance for other phases can be done. The design speed is chosen to be 27.6 m/s and required thrust for this phase is chosen to be 190 N. The thrust chosen is higher than the calculated value for loiter, mainly due to two factors. The propeller is chosen to be designed at the $(T/W)_{min}$ of 0.117, which requires approximately 190 N. Another reason is that the drag estimated in the analysis is still a crude approximation which might have under-predicted the drag and hence a factor of approximately 2 is applied in order to ensure that the UAV can maintain its loiter with an η_P of 0.8. Finally, while cruising and climbing, the thrust requirement will increase, and hence a minimum thrust of 190 N has been set for the design at this point.

4 Design Considerations of Parameters

4.1 Tip speeds and non-dimensional quantities

Non-dimensional analysis of Reynolds number and Mach number profile along the blade span is required for selecting aerofoils with high M_{DD} as the maximum allowed tip Mach number would be limited by compressibility effects at high RPM forward flight, which will be used in the blade optimisation. This can also be used to predict structural integrity. Initially, estimated properties including blade radius, hub radius, chord distribution and rotational rates, have been selected based on similar propellers used in [4] and [5]. The initial predicted Reynolds number and Mach number profiles are shown in Fig. 8. The tip speed reaches 104.7 m/s, if RPM of 2000 is used, with

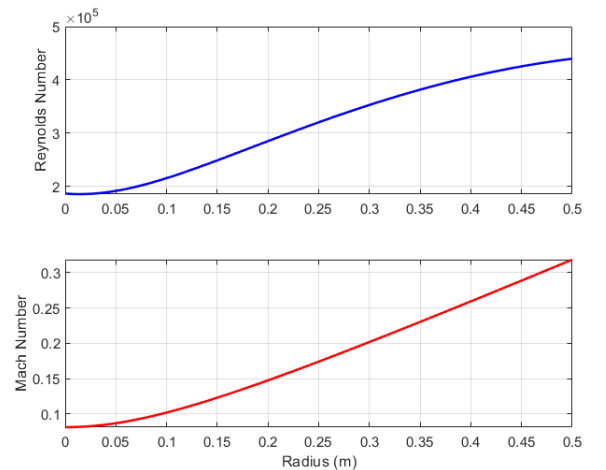


Figure 8. Estimated Reynolds number and Mach number distribution along a single propeller blade

a tip Mach number of 0.317, which needs to be below the tip drag-divergence number, such that stable conditions and lower wave drag is achieved. The design Reynolds number can be selected by looking at the maximum Re in Fig. 8, which is around 4.5×10^5 , and hence the aerofoils can be shortlisted based on this value.

4.2 Tip Diameter

Estimating tip diameter is crucial as it would determine various characteristics such as thrust for performance, geometric constraints, such as ground-clearance, and insights on propeller noise. Excessively large diameter might lead to too high tip speeds with noise, and propeller strike issues [3]. The tip diameter required is estimated using empirical model set out by Stinton [11] for the given loiter speed, V_{loiter} , maximum engine power, P_{BHP} in HP, and RPM of 2500, which is common RPM of propellers [8], by using eq.(1). It is deduced that all of these n-bladed propellers, where $2 \leq n \leq 4$, required a diameter greater than 1 m, which violates the constraint defined initially. Hence, an analysis on increasing the number of blades is required to find the effects on the blade diameter.

Table 3. Initial propeller tip diameter estimation

No. of blades	Tip Diameter(m)
2	1.45
3	1.33
4	1.21

$$D = 254 \left[\left(\frac{P_{BHP}}{53.5 \cdot \text{RPM}^2 \cdot V_{loiter} \cdot 16} \right)^{0.25} + \left(\frac{P_{BHP}}{111 \cdot \text{RPM}^2 \cdot V_{loiter} \cdot 16} \right)^{0.25} \right] \quad (1)$$

4.3 Number of blades

Increasing the number of blades raises the torque needed to turn the propeller, considering every other parameter is kept constant. This will lead to a reduced power available per blade, and hence the UAV performance will suffer. Hence, when increasing the number of blades, the diameter has to reduce, such that the torque generated per blade is reduced, which allows the propeller to swing at the same RPM as original one. This further helps in reduction of weight per blade, ground clearance constraint, propeller noise, etc. To estimate the reduction in diameter needed when increasing the number of blades from 2 blades, with 1.45 m diameter, to 3, 4 and 5, eq. (2) [3] is iteratively solved for D_{new} . The aim is to minimise the diameter while also using the minimum number of blades required.

$$V_{loiter}^2 D_{new} + (\pi n k)^2 D_{new}^3 = (V_{loiter}^2 D_{old} + (\pi n k)^2 D_{old}^3) \left(\frac{N_{B_{old}}}{N_{B_{new}}} \right) \left(\frac{S_{old}}{S_{new}} \right) \quad (2)$$

where N_B is number of blades, S is the blade surface area, n is revolutions per second and k is fraction spanwise location of the center of pressure on the blade, which is typically (0.65 - 0.75) [3].

From fig.9a, it can be seen when increasing from 2 to 3 blade, the diameter reduces by approximately 21% in cruise, keeping other variables constant. The blade surface area reduction ratio required for this change will be approximately 0.9, so 10% reduction in the blade area, as shown in fig. 9b. A simplified model[3], eq. (3) for the diameter length change is also plotted to consider the area change shown in fig. 9b, which gives approximately similar results to the original equation. Due to the blanketing effects from the fuselage, if we choose 4 or more blades, the diameter becomes too small, and we would lose performance due to a reduction in propeller stream-tube area. Hence, the optimal number of blades selected is 3. This defines the final design tip diameter to be 0.88 m. Below 3-blades, the diameter would need to be larger than 1 m, which is not possible due to design constraints.

$$D_{new} = D_{old} \sqrt[3]{\left(\frac{N_{B_{old}}}{N_{B_{new}}} \right) \left(\frac{S_{old}}{S_{new}} \right)} \quad (3)$$

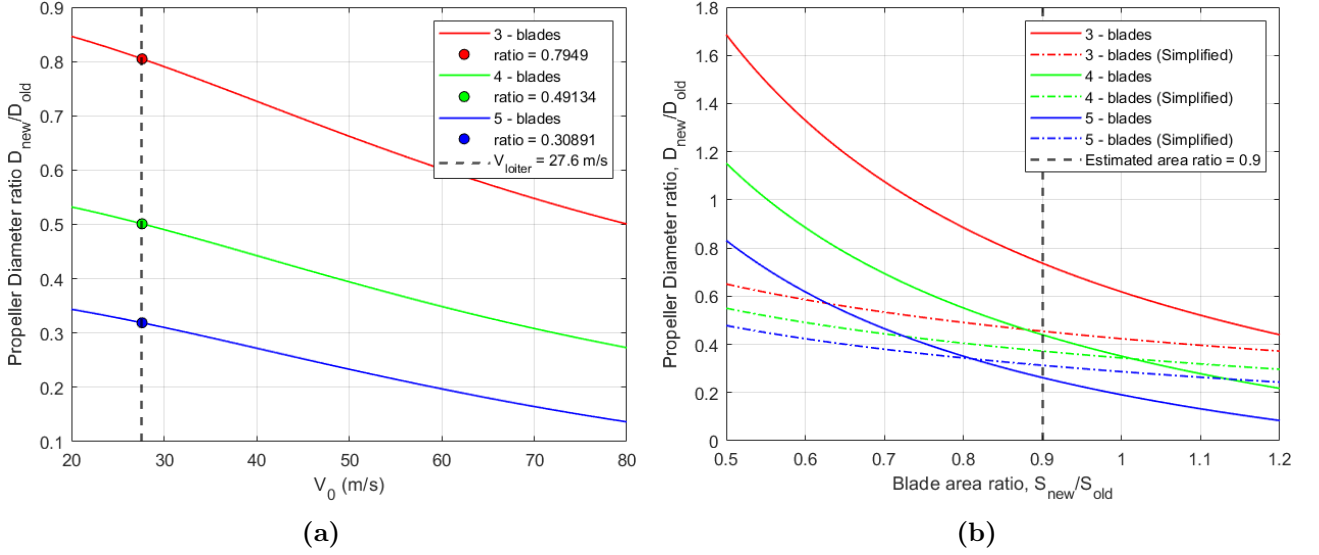


Figure 9. Effect of airspeed (a) and blade area ratio (b) on the size of a n-bladed ($n = 3, 4, 5$) propeller when replaced with 2-bladed propeller

4.4 Hub Radius

The presence of the hub has a large effect on the sectional lift distribution, as the lift tends to 0 at this point. This directly impacts the thrust produced by the propeller. It is important to consider this, when performing optimisation of blade design and this reduction in the lift generated needs to be approximated using an appropriate method. Then, the aim becomes to minimise these effects to reduce losses. The initial step to achieve this is to minimise the hub radius, while also keeping structural considerations. According to an article [12], the initial estimation of the hub radius must be 10-25% of the propeller tip radius, so that range to consider in the optimisation is 0.044 - 0.11 m.

4.5 Aspect Ratio

In BEMT, the thrust generated can be calculated once the radius, R , rotational rate n , and solidity σ are known. The blade solidity which is the ratio of blade area to the actuator disc area, is defined using eq. (4).

$$\sigma = \frac{N_B c R}{\pi R^2} \quad (4) \quad \mathcal{R} = \frac{(2R)^2}{2cR} = \frac{2N_B}{\pi\sigma} \quad (5)$$

where N_B is number of blades and c is chord length of blade (initially assumed to be constant). It is known that when the solidity is increased, the design point has to shift to larger advance ratios. The solidity also has a greater effect on the efficiency, as more disc area is captured by the blades [13]. The aspect ratio of the blade is also another equivalent way to represent the solidity, but more commonly used when considering aerodynamics. It can be related to solidity using eq. (5).

From the other parameters considered until now, the design $\mathcal{R}$ required is 8.8 with solidity σ of 0.22, and these values will be used to evaluate the final design. This will likely to change as more iterations in the optimisation are performed, when the local solidity will be affected based on the chord distribution.

4.6 Summary of the parameters

The design parameters considered until now have been finalised in table 4. These parameters will be used in the optimisation problem to achieve the correct target requirements.

Table 4. Target Design point

Parameter	Value	Considerations
Design speed	27.6 m/s	Loiter speed
Design thrust	190 N	$T = D$
Design Reynolds Number	500000	Max. along the blade
Design Mach Number	0.6	Max. at the tip
Rotational speed, RPM	1500 - 6500	changes in different phases
Number of blades, N_B	3	Stays constant in design
Propeller diameter, D	0.88	Stays constant in design
Hub radius	0.044 - 0.11 m	needs to be minimised
Solidity, σ	0.22	Varies along the blade(post optimisation)
Aspect ratio, $\mathcal{R}$	8.8	Target value

5 Optimisation of Propeller Design

5.1 Theory of optimisation

The main methods which can be used to design propeller blades include Blade Element Momentum Theory (BEMT), Actuator Disk theory, Vortex Line Theory, CFD, etc, with each using its own assumptions and equations to analyse the aerodynamic forces. Vortex line theory [14], requires an intensive numerical simulation as it analyses the tip vortex shedding in addition to modelling the propeller blade as a lifting line. CFD can be used to simulate the conditions and obtain performance metrics of designed propeller, however, optimising the blade geometry would require hybrid models which use genetic algorithms [15] to find the optimum points, which is very computationally expensive and time inefficient for the scope of this project. Therefore, BEMT method is selected for analysing the theory and optimising the propeller blade geometry, which divides the blade into a finite number of elements and individual elemental lift and drag contribution is integrated over the blade span to find overall thrust, torque and drag. Contrasting to actuator disk theory, it considers varying blade geometry, with variable chord distribution and aerodynamic characteristics[3]. The differential lift and drag forces of a single, dL and dD are given by eq. (6) as shown below.

$$dL = \frac{1}{2}\rho V^2 \cdot c \cdot C_l \cdot dr \quad dD = \frac{1}{2}\rho V^2 \cdot c \cdot C_d \cdot dr \quad (6)$$

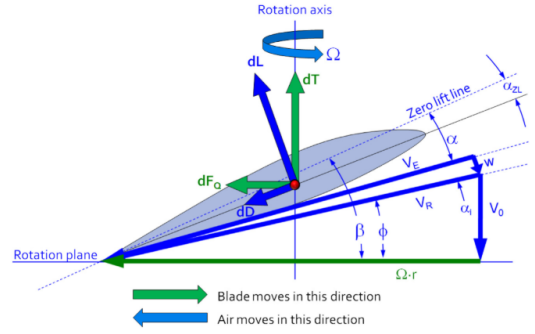
where $c(r)$ is the chord distribution, C_l and C_d are lift and drag coefficient of each individual blade element, respectively.

The thrust and torque generated by the propeller then can be calculated using the integral sum of these lift and drag components using the eq. (7) and (8)[3], respectively.

$$T = N_B \int_{R_{hub}}^R dT = N_B \int_{R_{hub}}^R dL \cos(\phi + \alpha_i) - N_B \int_{R_{hub}}^R dD \sin(\phi + \alpha_i) \quad (7)$$

$$Q = N_B \int_{R_{hub}}^R r \cdot dF_Q = N_B \int_{R_{hub}}^R r \cdot dL \sin(\phi + \alpha_i) + N_B \int_{R_{hub}}^R r \cdot dD \cos(\phi + \alpha_i) \quad (8)$$

where N_B is the number of blades, R is the tip radius and R_{hub} is the hub radius, ϕ is the flow angle, and α_i is the angle of incidence.

**Figure 10.** Forces, angles and velocity acting on the propeller blade[3]

There is an induced angle of attack, due to the w component shown in 10, which affects the thrust, which will be considered in the optimisation as induced velocity factors. In the optimisation, instead of evaluating these equations, they will be solved for given conditions through iterative process using numerical integration schemes, to work out the optimum chord and twist distribution.

5.2 Design methodology

Designing an optimal propeller geometry involves finding a complex array of parameters, where certain design constraints limits the optimisation potential such as constraints on tip diameter, Hub radius, and number of blades. Hence, a set method is chosen to develop a numerical code to find the optimal design of the propeller blade as shown in the Fig. 11. In this approach, different parameters are varied using fast iterations, to evaluate the power required for a given thrust required or vice versa. This approach is chosen because a complete analytical optimisation would be unfeasible due to the complexity of these parameters varying and it will often contradict some design considerations.

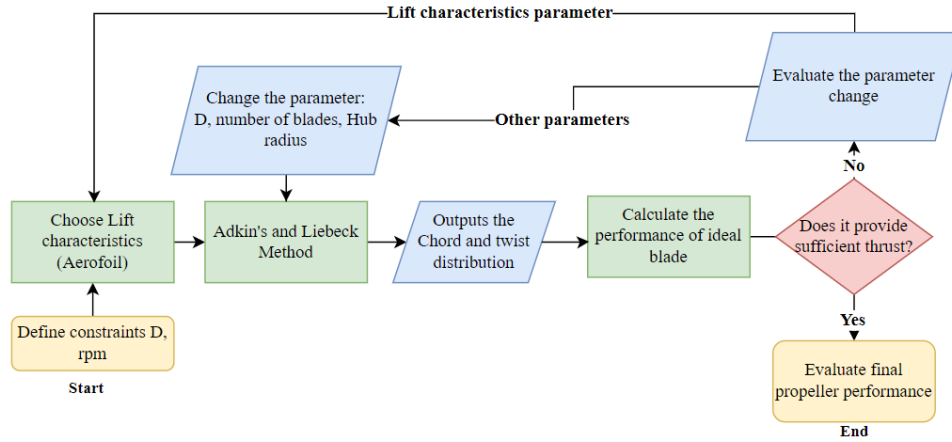


Figure 11. Overall design approach taken for optimum propeller design

5.3 Optimisation using Adkin's and Liebeck Method

Many conventional methods used for finding the optimal blade geometry involve initial estimation of axial and tangential induction factors, followed by iterative application of BEMT equations until convergence is achieved for the desired thrust. A similar methodology was introduced by Adkin and Liebeck[16].

Adkin and Liebeck aimed to refine and correct existing propeller design methods constructed by H. Glauret, Betz, Theodorsen and Larrabee [16], It began in 1936 with H. Glauret authoring a work on combining momentum theorem with BEMT, and correcting it for momentum losses due to radial flow, however the propeller wake was neglected. Then Betz found the conditions under which a design would have the energy loss minimised in 1919. He used small angle approximations, which limits the analysis to only light loading conditions. Theodorsen, proved that the Betz conditions can also be applied to heavy loading analysis. Following these improvements, the method still had some inaccuracies including small angle approximations, the displacement velocity solution being only valid for light loading and the analysis was done using inviscid assumptions, hence ignoring all the induced velocity terms. Adkin and Liebeck further corrected all these inaccuracies and finalised a set method to follow when designing a propeller, as shown in Fig. 12.

The method outlined by Adkin and Liebeck initiates with specified conditions of power into the propeller or thrust generated by propeller, hub radius, tip diameter, RPM, flight speed, number of blades and finite number of stations(elements), which calculates the blade geometry. Then, the method iterates until the displacement velocity factor ζ , converges to a value within a tolerance defined.

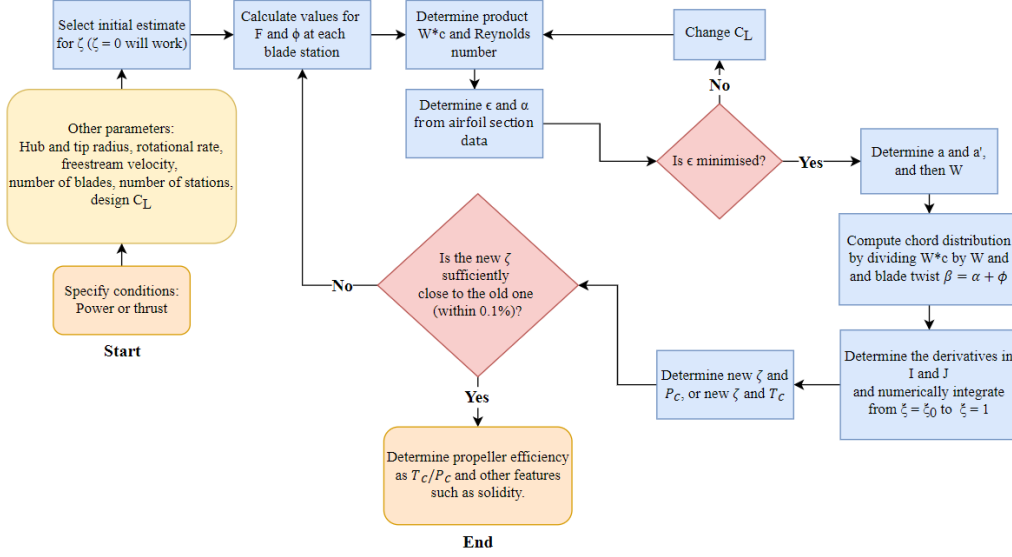


Figure 12. Adkin and Liebeck's optimisation method for propeller design

At each iteration, the method calculates, Prandtl momentum tip loss factor, F , flow angle, ϕ , the product of total velocity and local chord, Wc , and Reynolds number at each blade station using the equations given below.

$$F = \frac{2}{\pi} \arccos(e^{-f}) \quad (9)$$

$$f = \left(\frac{N_B}{2}\right) \left(\frac{1 - \xi}{\sin(\phi_t)}\right) \quad (10)$$

where, ϕ_t is the tip flow angle.

$$\tan(\phi_t) = \lambda \left(1 + \frac{\zeta}{2}\right) \quad (11)$$

$$\tan(\phi) = \frac{\tan(\phi_t)}{\xi} \quad (12)$$

where λ is speed ratio and ξ is non-dimensional radius.

$$Wc = \frac{4\pi\lambda GVR\zeta}{C_L N_B} \quad (13)$$

$$G = F \cos(\phi) \sin(\phi) \quad (14)$$

where C_L is the local design lift coefficient.

Once these steps are done, the method requires us to maximise the lift-to-drag, $\frac{1}{\epsilon}$ and provide the required α_{design} to achieve that ϵ . This data is obtained from airfoil tools website [17], which uses XFOIL analysis. Following this, the method then calculates the axial and tangential induction factors, where the condition for minimum energy losses results in these interference factors being related to ζ given by the equations below.

$$a = \frac{\zeta}{2} \cos^2(\phi) (1 - \epsilon \tan(\phi)) \quad (15) \quad a' = \frac{\zeta}{2\xi} \cos(\phi) \sin(\phi) \left(1 + \frac{\epsilon}{\tan(\phi)}\right) \quad (16)$$

Then the chord distribution is calculated by dividing the product Wc , by total velocity, W . Some further equations [16] and numerical integration techniques are used to calculate new ζ based on the geometry and compared against the old one, and this process repeats until we break a tolerance condition specified initially. Finally, the desired characteristic(power or thrust) is calculated and compared to design and constraint requirements. This method is implemented using MATLAB and the final code can be found in appendix A.1.

5.4 Aerofoil Selection

Selecting the candidate aerofoil for the blade required to shortlist the aerofoils based on the design requirements. For best aerodynamic performance, the aerofoil needs to produce more lift with minimised drag, hence a high $(L/D)_{max}$ is required. Consequently, the lift produced at this $(L/D)_{max}$ must be sufficient and as high as possible at the operating angle of attack. The aerofoil should perform well for a range of Reynolds number as it varies along the blade. Considering the structural integrity, the aerofoil thickness must be adequate, but not too high as it leads to increased drag. Taking all these points into account, 7 aerofoils are shortlisted using XFOIL data for $Re = 500000$ (closest to initial

estimated Re), (full data can be found in appendix C and then the aerofoil selection is integrated into the Adkin and Liebeck's Method, where after calculating the Reynolds number at each station along the blade, the aerodynamic characteristics such L/D , C_L , angle of attack, α , and M_{DD} , based on the Korn's model[18] and local c_L , shown in eq.(17) are updated according to the local Reynolds number. This is an iterative process and the main performance metric to compare the shortlisted aerofoils is the power required in loiter.

$$M_{DD} \cos \Lambda_{0.5} + \frac{C_l}{10 \cos^2 \Lambda_{0.5}} + \frac{t/c}{\cos \Lambda_{0.5}} = \kappa_\Lambda \quad (17)$$

where $\Lambda_{0.5}$ is the sweep angle(midchord), equal to 0, as no sweep has been considered for the blade, κ_Λ is the Korn factor = 0.87 [18], for conventional airfoils.

Table 5. Comparison of aerofoils considered for the propeller blade cross section

Aerofoil	$(t/c)_{max}$	$(L/D)_{max}$	C_L @ $(L/D)_{max}$	Power required in loiter (kW)
Clark-y	0.117	114.8	1.11	7.31
VR-12	0.106	94.4	1.19	7.37
SG-6043	0.1	175	1.03	7.18
AH-7476	0.059	138	1.21	7.19
GOE-804	0.06	211	1.04	7.14
E-216	0.104	155	1.02	7.16
NACA-0012-64	0.12	63	0.98	7.38

The aim here is to minimise the power required for loitering. It can be seen in table 5, GOE-804 aerofoil requires the least power in loiter, however, the $(t/c)_{max}$ is the smallest, and hence not viable option structurally. After this aerofoil, the least power requirement is given by E-216, with a reasonably high $(L/D)_{max}$ of 155 and $(t/c)_{max}$ of 10.4%, which is hence the best option. Other aerofoils such SG-6043 has quite high $(L/D)_{max}$, however it requires an increased power input and similarly Clark-y has quite good $(t/c)_{max}$, however increased power requirements, so they are not considered as efficient options.

5.5 Results

After performing the optimisation based on the loiter design point using the method explained, the results are presented. The optimum RPM for loiter is found to be 2000, which is deduced from a 3D map of efficiency against both speed and RPM, shown in appendix B.2, where the efficiency peaks at RPM equal at 2000 at velocity of 27.6 m/s. A similar plot is considered for hub radius, and it is found that hub radius doesn't significantly impacts the power required, however, the optimal hub radius is found to 0.06 m, as it provides a good balance of losses and structural integrity. The optimised Reynolds number and drag divergence Mach number distribution along the propeller blade is shown in Fig. 17, while also plotting the local Mach number distribution at the cruise speed, which is 41.1 m/s. It can be confirmed from these results that the stations along the blade will not reach the local M_{DD} in off-design conditions of both loiter and cruise,(since loiter speed < cruise speed). The Reynolds number distribution is quite close to the estimated one in initial stages, shown in Fig. 8, hence the initial ballpark values for parameters are found to be close to optimum values.

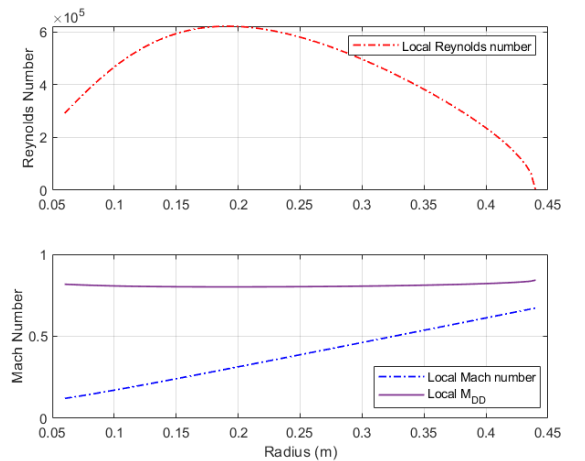


Figure 13. Optimised Reynolds number and Mach number distribution along a single propeller blade

The optimised chord and twist profile is shown below in Fig.14. These results are compared to Qblade (a software used for propeller optimisation), in order to validate the accuracy and correctness of the in-house MATLAB code. It is observed in the Fig.14 (a) that the distribution do not perfectly match. This can be explained by inaccuracies in Qblade modelling. Firstly, the application works on a constant Reynolds number, hence it affects the optimised chord distribution, which affects other results such as lift distribution and drag analysis. For the twist distribution, the results are quite comparable to each other, other than a approximately constant shift of 6° , which is again due to the inaccuracy in assuming the Reynolds number and $(L/D)_{max}$ constant, which creates a discrepancy in the optimisation results.

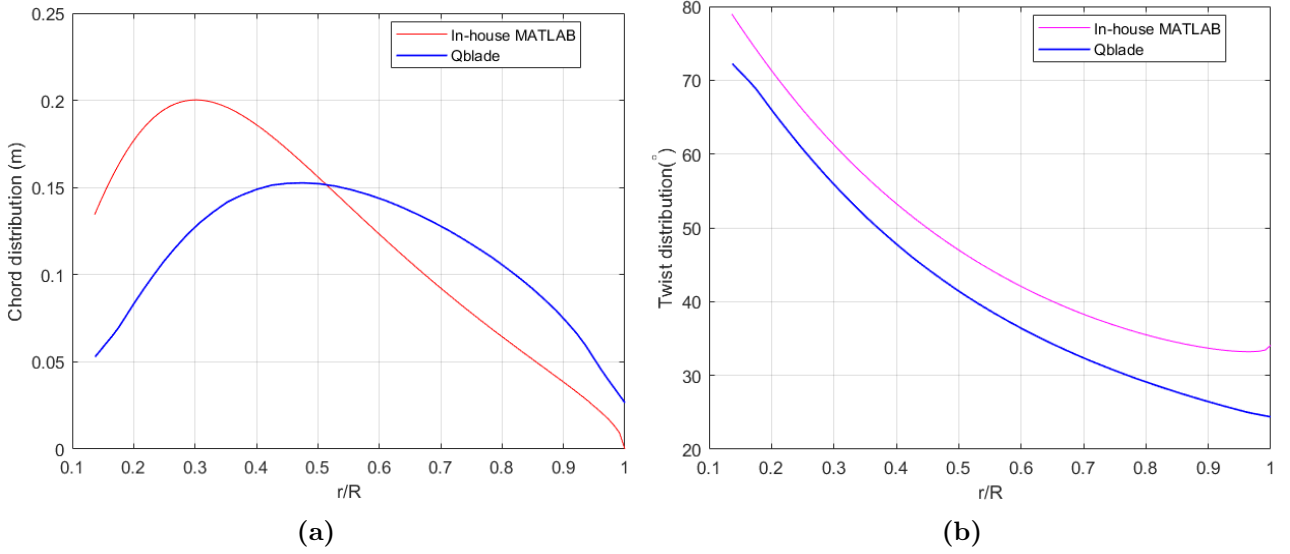


Figure 14. Optimised chord and twist distribution along a single propeller blade generated from both in-house MATLAB and Qblade, using Adkin and Liebeck’s Method

In the Fig. 25 given in appendix D, the optimised C_L distribution and the local angle of attack at each station is presented. It is found that the local AoA, α , is reasonably below the local stall AoA, which makes the blade suitable for off-design conditions, such that the blade will still have enough L/D , without stalling, which could have affected the UAV performance, if not considered, as the blade would have sudden lift loss, and consequently, loss in thrust produced.

6 Performance Analysis

6.1 Optimal Design and comparison with XROTOR

To test the validity of the BEMT and Adkin and Lieback method implementation in in-house MATLAB code, the results are compared against XROTOR, a program used for running BEMT analysis, developed by MIT[19]. Parameters are kept constant across both approaches, including RPM of 2000, loiter speed of 27.6 m/s, power input of 7 kW and the E-216 lift and drag polar data. The optimised geometry is inputted into XROTOR and BEMT results were produced. Detailed plotted results can be found in appendix E. The comparison of the key results has been tabulated in the table 6.

Table 6. Optimal design values comparison with XROTOR results

Design parameter	MATLAB	XROTOR result	% difference
Thrust (N)	190	177	6.8
Torque (Nm)	31.79	33.40	4.8
η_P	0.6121	0.6980	12

It is observed that there is a significant percentage difference in the quantities computed. While this difference is large, it can be accounted for by considering some factors affecting the results. Firstly, XROTOR does not account for variable Reynolds number along the blade, whereas the in-house code optimises the aerodynamic quantities such as $(L/D)_{max}$, C_L , angle of attack, etc based

on local Reynolds number, hence this could have caused an inconsistency. Secondly, XROTOR uses more complex and computationally intensive integration schemes for thrust calculation, whereas the in-house code uses trapezium rule, which is computationally efficient, to sum up each blade element contribution. This could have also affected the results. Even after these differences, the results are relatively close to each other, and hence it can be deduced that optimised geometry will perform effectively for the purpose of this mission.

6.2 Performance testing using Qblade

Performance testing of the optimised blade geometry is performed using QBlade software to make the BEMT predictions for different phases and off-design conditions of the flight more efficiently, complying to project time constraints. Fig.15 presents the propeller performance in terms of thrust/power coefficients and propeller efficiency, which is obtained using equations provided in appendix F for loiter(left) and cruise(right) simulations performed for increasing RPM from 1500 to 6500, which are the limits allowed by the engine. It can be deduced from the plot (a) that the η_P is very close to 0.8, which validates the designed geometry for loiter. Also, in plot (b), for an RPM of 2000, η_P of 0.78 is achieved in cruise, with an advance ratio of 1.503, which is a reasonable reduction of 2.5% from design value of 0.8, that can be accounted from losses increasing at higher speeds and can be tackled by variable power input.

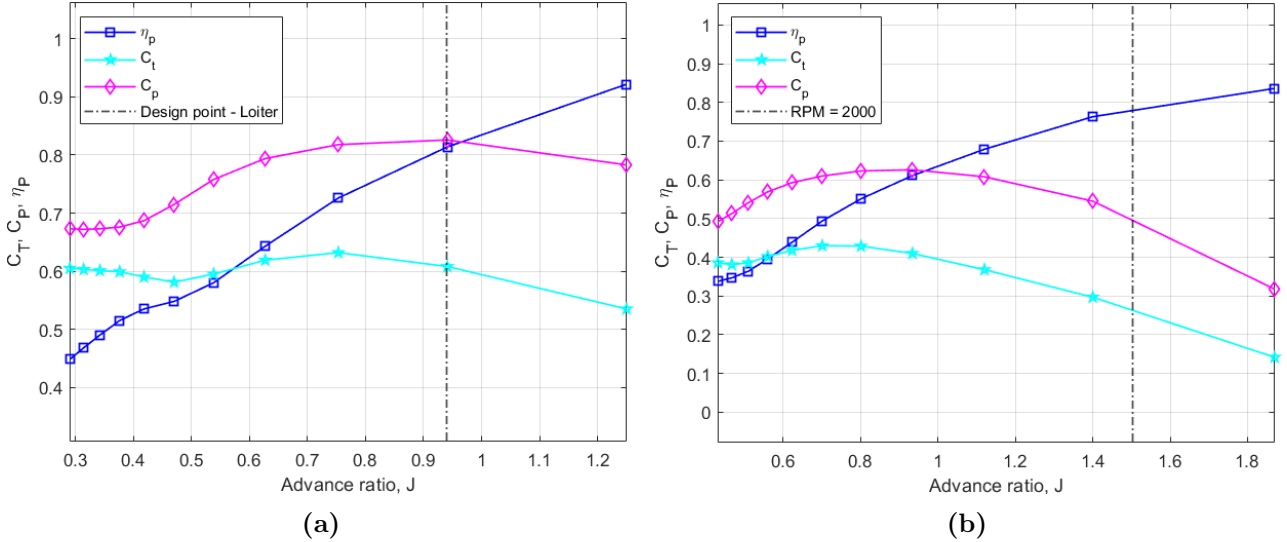


Figure 15. Propeller performance coefficients against advance ratio, for loiter(a) and cruise speed(b)

6.3 Results from CFD simulations

To further validate the BEMT predictions, CFD simulations have been performed using an ideal and unsteady model setup, with $k - \omega$ turbulence conditions. Two meshes were used in the simulations: fluid mesh and rotating mesh, with minimum surface size of fluid and rotating mesh equal to 0.5 cm and 0.1 cm, respectively, such that the rotating mesh is more finer than fluid, to capture better data. Detailed explanation of the CFD simulations can be found in Matthew's report[20]. The loiter simulation data for thrust and torque generated is presented in Fig.16, which compares BEMT predictions to CFD data for with and without fuselage geometries. Overall, the CFD results follow very close trend to BEMT predictions, though there is still a deviation present. To quantify this difference in data, the root mean square error is calculated. For thrust predictions, there is ± 16.9 N and ± 38.9 N error for CFD with fuselage and without fuselage, respectively, compared against BEMT. For torque predictions, there is ± 10.1 Nm and ± 8.3 Nm error for CFD with fuselage and without fuselage, respectively. The deviation between the BEMT and CFD can be explained using some key differences present in both methods. CFD simulates viscosity, turbulence and wake of the propeller blades when computing the values, whereas BEMT neglects these conditions. The CFD uses compressible aerodynamics, which considers density changes in the flow, whereas BEMT doesn't.

There are hub losses included in the CFD simulations, whereas BEMT does not consider the presence of hub. All these factors, explain the CFD results without fuselage being lower than BEMT, which explains that BEMT over-predicts performance conditions. Considering the error is smaller for fuselage added simulations than without the fuselage, suggests that the flow conditions with fuselage provide better efficiency, which can be explained using adverse pressure gradient towards the end of fuselage, which causes the flow to accelerate, and causes the total local velocity to increase.

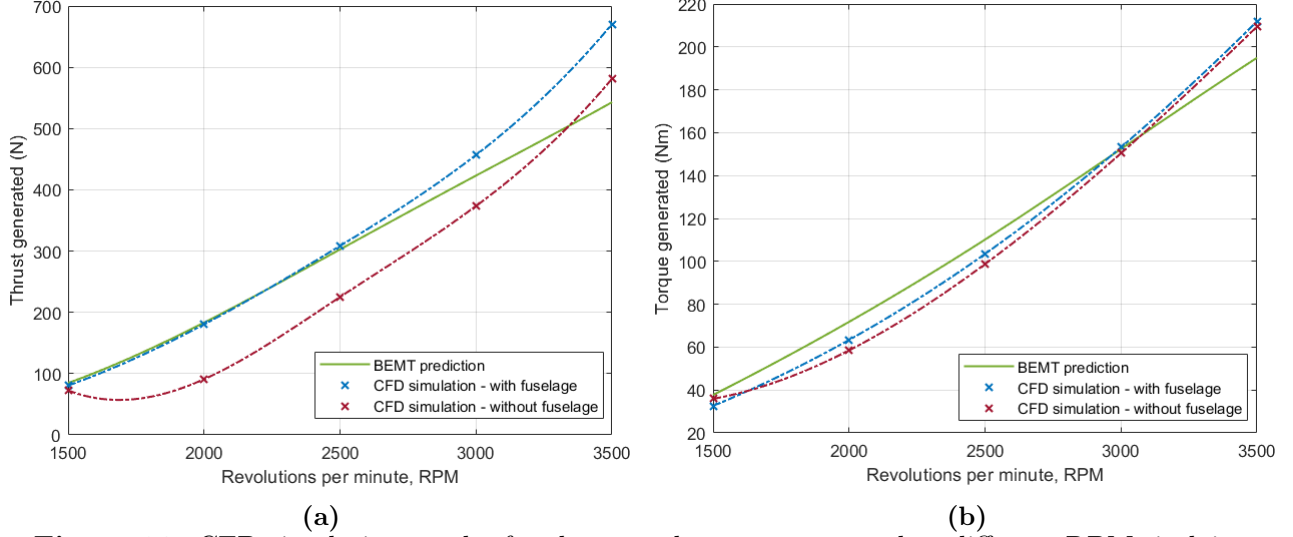


Figure 16. CFD simulation results for thrust and torque generated at different RPMs in loiter, compared to BEMT predictions

7 Conclusion and Discussion

To summarise, the report provides a detailed method to optimise the blade geometry of the pusher propeller which will be the drive-train of the UAV designed for disaster relief. First iteration of the optimisation has been done using Adkin and Liebeck method which utilises BEMT, and the results have been compared with multiple sources, including XROTOR, QBlade, and CFD. The results from different sources are within accepted degree of accuracy to BEMT, which validates overall design. The optimisation concentrates on finding the optimum chord and twist distribution for a given power/thrust and flight speed, subject to a number of constraints, by minimising the energy losses. Under this optimisation, a power input of 7.16 kW is required in the loiter phase with thrust produced of 190 N, that provides a propeller efficiency of 0.8 which was chosen as the design point. This is only 40% of full throttle for the selected engine, and only 2.2% higher than the initial design target power required, which confirms the optimality of this design phase. Hence, the design can be confirmed as suitable for the requirements, which were defined and justified before.

7.1 Further improvements

The optimised propeller iteration and prototypes needs to be tested in wind tunnel to validate all the sources of data, including BEMT. The propeller design can be optimised further in next iteration by considering some complex changes. By adjusting for different aerodynamic loads, using different aerofoils for the inboard and outboard sections can improve performance and structural integrity. The use of ISight's (an optimisation software) integration with MATLAB will enable an effective fine-tuning of design parameters through the use of complex optimisation algorithms like genetic algorithms. A more aerodynamic hub design can reduce hub losses and increase overall efficiency. Additional CFD simulations will offer better understanding of performance and limitations under off-design conditions. The propeller blade wake can be predicted more precisely by using the free wake vortex lattice method. Optimising the propeller's performance also requires rewriting the code to incorporate specific parameters such as hub geometry, structural data, materials, and viscosity/density changes.

References

- [1] Tharm Chursuk. Halo reach: Optimising weight and balance through comprehensive powerplant sizing for vtol fixed-wing uav. GDP report 02048996, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [2] UAVHE. Rw1-79, 2023. URL <https://uavhe.eu/products/rw1-79/>. Accessed: 2024-06-14.
- [3] Snorri. Gudmundsson. *General aviation aircraft design : applied methods and procedures*. Gale eBooks. Butterworth-Heinemann, Oxford, first edition. edition, 2014. ISBN 0-12-809998-4.
- [4] Mike Ball. Elevonx develops long-range fixed-wing hybrid vtol uavs. *Unmanned Systems Technology*, April 2021. URL <https://www.unmannedsystemstechnology.com/2021/04/elevonx-develops-long-range-fixed-wing-hybrid-vtol-uavs/>.
- [5] BlueBird Aero Systems. Thunderb-vtol, 2021. URL <https://bluebird-uav.com/thunderb-vtol/>. Accessed: 2024-06-14.
- [6] Zhiyan Wang. Halo reach: Material, manufacturing, undercarriage, boom and dynamics. GDP report 01911209, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [7] Jiaxun (Aaron) Leng. Halo reach: V-n diagram, tail, rotor and propeller loads and structural design. GDP report 02015060, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [8] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, Inc, 2018.
- [9] Inc. EPI. Selecting a propeller. https://www.epi-eng.com/propeller_technology/selecting_a_propeller.htm, 2024. Accessed: 2024-06-14.
- [10] Max F Lawlor. Halo reach: Drag analysis methodology and validity. GDP report 02026199, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [11] D. Stinton. *The Design of the Aeroplane*. Collins Professional and Technical Books, 1998.
- [12] PropellerPages. Propeller terminology - design considerations. http://www.propellerpages.com/?c=tech_info&f=terminology. Accessed: 2024-06-11.
- [13] Alfred Gessow and Garry Myers. *Aerodynamics of the Helicopter*. Frederick Ungar Publishing, New York, NY, 8th edition, 1985.
- [14] X. Zheng, X. Wang, Z. Cheng, and D. Han. The efficiency analysis of high-altitude propeller based on vortex lattice lifting line theory. *The Aeronautical Journal*, 121(1236):141–162, 2017. doi: 10.1017/aer.2016.112.
- [15] I. Cayiroglu and R. Kilic. Wing aerodynamic optimization by using genetic algorithm and ansys. *Acta Physica Polonica A*, 132(3-II):981–985, September 2017.
- [16] Charles N. Adkins and Robert H. Liebeck. Design of optimum propellers. *Journal of Propulsion and Power*, 10(5):676–682, 1994. doi: 10.2514/3.23779. URL <https://doi.org/10.2514/3.23779>.
- [17] Airfoil Tools. Airfoil tools. <http://www.airfoiltools.com/>. Accessed: 2024-06-15.
- [18] Ohad Gur, William H. Mason, and Joseph A. Schetz. Full-configuration drag estimation. *Journal of Aircraft*, 47(4):1356–1367, 2010. doi: 10.2514/1.47557. URL <https://doi.org/10.2514/1.47557>.
- [19] Mark Drela. MIT xrotor software. <https://web.mit.edu/drela/Public/web/xrotor/>. Accessed: 2024-06-15.
- [20] Matthew M Homa. Halo reach: Wing design and computational propeller analysis. GDP report 02038403, Dept. of Aeronautical Engineering, Imperial College London, 2024.

Appendix

A MATLAB code

A.1 Adkin's and Liebeck's implementation

Note that Adkin's and Liebeck's optimisation method is converted into a MATLAB function so that multiple iterations can be run using the same code.

```
1 function [chord, twist, alpha_rad, epsilon, zeta_new, eta_p,
   output_value, CL, C_T, C_P, prop_efficiency, stall_angle, Reynolds
   , M_dd] = Adkins(Hub_r, R, RPM, V, B, stations, C_L, alpha, epsilon
   , option, value)
2
3 % Define the given parameters
4 rho = 1.225; % Air density
5 Hub_r = Hub_r; % Hub radius %
6 R = R; % Tip radius %
7 RPM = RPM;
8 omega = RPM * 2 * pi / 60; % Rotational rate/omega %
9 V = V; % Freestream velocity
10 B = B; % Number of blades %
11 stations = stations; % Finite number of stations
12 CL = C_L; % Design lift coefficient for each station %
13
14
15 %-----THIS IS AEROFOIL DATA OBTAINED FROM
16 %XFOIL-----
17 % %Clark - y
18 % Re = [50000, 100000, 200000, 500000, 1000000]; % Re's for
   aerofoil
19 % C_L = [1.2442, 1.0873, 0.9005, 0.8090, 1.1090];
20 % aoa = [9.25, 6.75, 4.75, 3.75, 6.75]; % aoa in degrees
21 % L_D = [29.6, 53, 73.2, 98.7, 114.8];
22 % C_L_Max = [1.3370, 1.3694, 1.3965, 1.4329, 1.5240];
23 % stall = [11.25, 12.5, 12.5, 13.5, 14.75];
24
25 % %VR-12
26 % Re = [50000, 100000, 200000, 500000, 1000000];
27 % C_L = [0.6680, 0.5755, 0.8721, 0.9869, 1.1959];
28 % aoa = [4, 3, 6.25, 7.25, 9.25];
29 % L_D = [28.9, 39.4, 49.2, 73, 94.4];
30 % C_L_Max = [1.0038, 1.2551, 1.3816, 1.4935, 1.5988];
31 % stall = [8.5, 12, 14.25, 15.75, 17.0];
32 %
33 % %SG-6043
34 % Re = [50000, 100000, 200000, 500000, 1000000];
35 % C_L = [1.4379, 1.3732, 1.2878, 1.1238, 1.0304];
36 % aoa = [8.75, 7, 5.5, 3.5, 2.5];
37 % L_D = [39.7, 66.5, 98, 143, 175.2];
38 % C_L_Max = [1.6133, 1.6491, 1.6277, 1.7038, 1.8311];
39 % stall = [12, 14.5, 16.75, 15.5, 17.5];
40 %
41 % %AH-7476
```

```

42 % Re = [50000, 100000, 200000, 500000, 1000000];
43 % C_L = [1.4225, 1.3188, 1.2215, 1.1100, 1.2057];
44 % aoa = [6.5, 5, 3.75, 2.5, 3.5];
45 % L_D = [48.1, 75.7, 107.2, 148.4, 138.1];
46 % C_L_Max = [1.5981, 1.6180, 1.6263, 1.6596, 1.7337];
47 % stall = [10.25, 10.75, 11, 12, 13];
48 %
49 % %GOE-804
50 % Re = [50000, 100000, 200000, 500000, 1000000];
51 % C_L = [1.2488, 1.2713, 1.2615, 1.1415, 1.0441];
52 % aoa = [6.75, 5.75, 4.75, 3, 1.75];
53 % L_D = [43.7, 70.6, 107.1, 169.1, 211.4];
54 % C_L_Max = [1.4969, 1.5121, 1.5626, 1.6123, 1.7161];
55 % stall = [9, 9.25, 9.5, 10.75, 12.5];
56 %
57 % %E-216
58 Re = [50000, 100000, 200000, 500000, 1000000];
59 C_L = [1.4618, 1.3516, 1.1901, 1.1344, 1.0159];
60 aoa = [8, 6, 4, 3.25, 1.75];
61 L_D = [40.7, 68.5, 99.2, 136.8, 155.1];
62 C_L_Max = [1.56, 1.5165, 1.5575, 1.6634, 1.7600];
63 stall = [10.25, 10.0, 10.25, 12.25, 13.25];
64 %
65 % %NACA-0012-64
66 % Re = [50000, 100000, 200000, 500000, 1000000];
67 % C_L = [0.5898, 0.5815, 0.5339, 0.8741, 0.9796];
68 % aoa = [4.5, 3.75, 3.25, 7.5, 9];
69 % L_D = [26.7, 37.8, 43.8, 50.8, 63];
70 % C_L_Max = [0.8179, 0.8935, 0.9536, 1.1204, 1.3431];
71 % stall = [8, 9, 9.75, 13.75, 16.5];
72
73 epsilon = 1./L_D;
74
75 % Fit polynomials (using a second-order polynomial as an example)
76 p_CL = polyfit(Re, C_L, 2);
77 p_aoa = polyfit(Re, aoa, 2);
78 p_LD = polyfit(Re, L_D, 2);
79 p_epsilon = polyfit(Re, epsilon, 2);
80 p_stall = polyfit(Re, stall, 2);
81
82 % Requirements %
83 if option == 'T'
84     T = value; % Thrust
85     %T_c = T / (rho * ((RPM / 60) ^ 2) * (R * 2) ^ 4);
86     T_c = 2 * T / (rho * (V^2) * pi * (R^2)); % Thrust coefficient
87 elseif option == 'P'
88     P = value * 0.8; % Power into the propeller
89     %P_c = P / (rho * ((RPM / 60) ^ 3) * (R * 2) ^ 5);
90     P_c = 2 * P / (rho * (V^3) * pi * (R^2)); % Power coefficient
91 end
92
93 % Define other constants

```

```

94 % Step 1: Initial guess for zeta
95 zeta = 0; % Initial guess for zeta
96 tolerance = 0.001; % 0.1% tolerance
97 max_iterations = 1000; % Maximum number of iterations to prevent
    infinite loop
98 iteration = 0; % Initialize iteration counter
99
100 % Initialize variables
101 zeta_new = 0.1; % Ensure the loop starts
102 while abs(zeta_new - zeta) / zeta_new > tolerance && iteration <
    max_iterations
103     zeta = zeta_new;
104
105     % Step 2: Calculate phi and F for each radial station
106     r = linspace(Hub_r, R, stations);
107
108     xi = zeros(1, stations);
109     phi = zeros(1, stations);
110     f = zeros(1, stations);
111     F = zeros(1, stations);
112     G = zeros(1, stations);
113     Wt = zeros(1, stations);
114     Wc = zeros(1, stations);
115     a = zeros(1, stations);
116     a_prime = zeros(1, stations);
117     W = zeros(1, stations);
118     chord = zeros(1, stations);
119     beta = zeros(1, stations);
120
121     for i = 1:stations
122         xi(i) = r(i) / R; % Non-dimensional radius = lambda * x
123     end
124
125     lambda = V / (omega * R); % Speed ratio V/omega*R
126
127     phi_t = atan(V / (omega * R) * (1 + zeta / 2)); % Flow angle
        at the tip
128
129     for i = 1:stations
130         phi(i) = atan((tan(phi_t)) / xi(i)); % Flow angle at each
            station
131     end
132
133     for i = 1:stations
134         f(i) = (B / 2) * (1 - xi(i)) / sin(phi_t);
135         F(i) = (2 / pi) * acos(exp(-f(i))); % Prandtl's tip loss
            factor
136     end
137     F(end) = 0;
138
139     % Step 3: Determine the product Wc
140     for i = 1:stations

```

```

141     Wt(i) = V * zeta * sin(phi(i)) * cos(phi(i)); %
        Tangential velocity
142     G(i) = F(i) * cos(phi(i)) * sin(phi(i)); % Circulation
        function
143     Gamma = 2 * pi * (V ^ 2) * G(i) / (B * omega); %
        Circulation
144     x = omega * r(i) / V; % Speed ratio
145
146     Wc(i) = 4 * pi * lambda * V * R * zeta * G(i) / (Cl * B);
147     kinematic_viscosity = 1.48 * 10^(-5);
148     Reynolds_number = Wc(i) / kinematic_viscosity;
149     CL(i) = 7;
150
151     %-----Aerodynamic optimisation using AEROFOIL
152     %DATA-----
153     while (abs(CL(i) - Cl) > 0.01) % Tolerance condition for
        local C_L
154         Cl = CL(i);
155         Wc(i) = 4 * pi * lambda * V * R * zeta * G(i) / (CL(i)
            * B);
156         Reynolds_number = Wc(i) / kinematic_viscosity;
157         CL(i) = polyval(p_CL, Reynolds_number); % Updating
            local C_L based on local Reynolds number
158     end
159
160     Reynolds(i) = Reynolds_number;
161
162     K_a = 0.8;
163     t_over_c = 0.103;
164
165     M_dd(i) = K_a - t_over_c + 0.1 * CL(i);
166
167     % Step 4: Determine epsilon and alpha_design
168     alpha_rad(i) = polyval(p_aoa, Reynolds_number) * pi / 180;
        % Angle of attack in degrees
169     epsilon(i) = polyval(p_epsilon, Reynolds_number); % Drag-
        to-lift ratio
170     stall_angle(i) = polyval(p_stall, Reynolds_number);
171
172     % Step 6: Determine a, a' and W
173     a(i) = (zeta / 2) * cos(phi(i))^2 * (1 - epsilon(i) * tan(
        phi(i))); % Axial interference factor
174     a_prime(i) = (zeta / (2 * x)) * cos(phi(i)) * sin(phi(i))
        * (1 + epsilon(i) / tan(phi(i))); % Rotational
        interference factor
175     W(i) = V * (1 + a(i)) / sin(phi(i)); % Local total
        velocity
176
177     % Step 7: Compute the chord and the blade twist
178     chord(i) = Wc(i) / W(i); % Chord calculation based on Wc
179     beta(i) = alpha_rad(i) + phi(i); % Twist distribution
180 end

```

```

181
182     % Step 8: Calculate derivatives - summing up all the lift and
183     drag
184     % contributions per element
185     I_prime_1 = zeros(1, stations);
186     I_prime_2 = zeros(1, stations);
187     J_prime_1 = zeros(1, stations);
188     J_prime_2 = zeros(1, stations);
189
190     for i = 1:stations
191         I_prime_1(i) = 4 * xi(i) * G(i) * (1 - epsilon(i) * tan(
192             phi(i)));
193         I_prime_2(i) = lambda * (I_prime_1(i) / (2 * xi(i))) * (1
194             + epsilon(i) / tan(phi(i))) * sin(phi(i)) * cos(phi(i))
195             ;
196         J_prime_1(i) = 4 * xi(i) * G(i) * (1 + epsilon(i) / tan(
197             phi(i)));
198         J_prime_2(i) = (J_prime_1(i) / 2) * (1 - epsilon(i) * tan(
199             phi(i))) * cos(phi(i))^2;
200     end
201
202     % Integrate the derivatives over xi using the trapezoidal rule
203     I_1 = trapz(xi, I_prime_1);
204     I_2 = trapz(xi, I_prime_2);
205     J_1 = trapz(xi, J_prime_1);
206     J_2 = trapz(xi, J_prime_2);
207
208     % Step 9: Determine zeta_new based on provided input
209     if exist('T_c', 'var')
210         % Use the first set of equations if T_c is provided
211         zeta_new = (I_1 / (2 * I_2)) - sqrt((I_1 / (2 * I_2))^2 -
212             T_c / I_2);
213         P_c = J_1 * zeta_new + J_2 * zeta_new^2;
214     elseif exist('P_c', 'var')
215         % Use the second set of equations if P_c is provided
216         zeta_new = -(J_1 / (2 * J_2)) + sqrt((J_1 / (2 * J_2))^2 +
217             P_c / J_2);
218         T_c = I_1 * zeta_new - I_2 * zeta_new^2;
219     else
220         error('Please provide either T_c or P_c as an input.');

```

```

224     eta_p = T_c / P_c; % Propeller efficiency - according to design
        procedure
225
226     if option == 'T'
227         %P = P_c * (rho * ((RPM / 60) ^ 3) * (R * 2) ^ 5);
228         P = P_c * (rho * (V^3) * pi * (R^2)) / 2;
229         output_value = P;
230     elseif option == 'P'
231         %T = T_c * (rho * ((RPM / 60) ^ 2) * (R * 2) ^ 4);
232         T = T_c * (rho * (V^2) * pi * (R^2)) / 2;
233         output_value = T;
234     end
235
236     C_T = T / (((rho * (RPM / 60) ^ 2) * (2 * R) ^ 4)); % Thrust
        coefficient
237     C_P = P / (((rho * (RPM / 60) ^ 3) * (2 * R) ^ 5)); % Power
        coefficient
238
239     prop_efficiency = (C_T / C_P) * (V / ((RPM / 60) * 2 * R)); %
        Propeller efficiency
240
241     % Return the twist distribution
242     twist = beta;
243 end

```

A.2 Performance code for fixed geometry input

```

1 function [output_value, prop_efficiency, C_T, C_P, C_Q, Q] = propeller
    (V, RPM, aoa_given, epsilon_given, twist_given, zeta, option, value
    )
2
3     % Define other constants
4     % Step 1: Initialize variables
5     Hub_r = 0.06;
6     R = 0.44;
7     stations = 100;
8     B = 3;
9     rho = 1.225;
10
11     omega = RPM * 2 * pi / 60;
12
13     r = linspace(Hub_r, R, stations);
14     xi = zeros(1, stations);
15     phi = zeros(1, stations);
16     f = zeros(1, stations);
17     F = zeros(1, stations);
18     G = zeros(1, stations);
19     Wt = zeros(1, stations);
20     Wc = zeros(1, stations);
21     a = zeros(1, stations);
22     a_prime = zeros(1, stations);
23     W = zeros(1, stations);

```

```

24     epsilon = zeros(1, stations);
25
26     % Requirements %
27     if option == 'T'
28         T = value; % Thrust
29         %T_c = T / (rho * ((RPM / 60) ^ 2) * (R * 2) ^ 4);
30         T_c = 2 * T / (rho * (V^2) * pi * (R^2)); % Thrust coefficient
31     elseif option == 'P'
32         P = value * 0.8; % Power into the propeller
33         %P_c = P / (rho * ((RPM / 60) ^ 3) * (R * 2) ^ 5);
34         P_c = 2 * P / (rho * (V^3) * pi * (R^2)); % Power coefficient
35     end
36
37     for i = 1:stations
38         xi(i) = r(i) / R; % Non-dimensional radius = lambda * x
39     end
40
41     lambda = V / (omega * R); % Speed ratio V/omega*R
42
43     for i = 1:stations
44         phi(i) = twist_given(i) - aoa_given(i); % Flow angle at each
            station
45     end
46
47     for i = 1:stations
48         f(i) = (B / 2) * (1 - xi(i)) / sin(phi(end));
49         F(i) = (2 / pi) * acos(exp(-f(i))); % Prandtl's tip loss
            factor
50     end
51     F(end) = 0;
52
53     % Step 3: Determine the product Wc
54     for i = 1:stations
55         G(i) = F(i) * cos(phi(i)) * sin(phi(i)); % Circulation
            function
56         %Gamma missing
57         x = omega * r(i) / V; % Speed ratio
58
59         epsilon(i) = epsilon_given(i); % Drag-to-lift ratio
60
61         % Step 6: Determine a, a' and W
62         a(i) = (zeta / 2) * cos(phi(i))^2 * (1 - epsilon(i) * tan(phi(
            i))); % Axial interference factor
63         a_prime(i) = (zeta / (2 * x)) * cos(phi(i)) * sin(phi(i)) * (1
            + epsilon(i) / tan(phi(i))); % Rotational interference
            factor
64         W(i) = V * (1 + a(i)) / sin(phi(i)); % Local total velocity
65     end
66
67     % Step 8: Calculate derivatives
68     I_prime_1 = zeros(1, stations);
69     I_prime_2 = zeros(1, stations);

```

```

70 J_prime_1 = zeros(1, stations);
71 J_prime_2 = zeros(1, stations);
72
73 for i = 1:stations
74     I_prime_1(i) = 4 * xi(i) * G(i) * (1 - epsilon(i) * tan(phi(i)
75         ));
76     I_prime_2(i) = lambda * (I_prime_1(i) / (2 * xi(i))) * (1 +
77         epsilon(i) / tan(phi(i))) * sin(phi(i)) * cos(phi(i));
78     J_prime_1(i) = 4 * xi(i) * G(i) * (1 + epsilon(i) / tan(phi(i)
79         ));
80     J_prime_2(i) = (J_prime_1(i) / 2) * (1 - epsilon(i) * tan(phi(
81         i))) * cos(phi(i))^2;
82 end
83
84 % Integrate the derivatives over xi using the trapezoidal rule
85 I_1 = trapz(xi, I_prime_1);
86 I_2 = trapz(xi, I_prime_2);
87 J_1 = trapz(xi, J_prime_1);
88 J_2 = trapz(xi, J_prime_2);
89
90 % Step 9: Determine zeta_new based on provided input
91 if exist('T_c', 'var')
92     % Use the first set of equations if T_c is provided
93     %zeta_new = (I_1 / (2 * I_2)) - sqrt((I_1 / (2 * I_2))^2 - T_c
94         / I_2);
95     P_c = J_1 * zeta + J_2 * zeta^2;
96 elseif exist('P_c', 'var')
97     % Use the second set of equations if P_c is provided
98     %zeta_new = -(J_1 / (2 * J_2)) + sqrt((J_1 / (2 * J_2))^2 +
99         P_c / J_2);
100     T_c = I_1 * zeta - I_2 * zeta^2;
101 else
102     error('Please provide either T_c or P_c as an input.');

```

```

113 C_P = P / (rho * (RPM / 60) ^ 3 * (2 * R) ^ 5); % Power
      coefficient
114
115 prop_efficiency = (C_T / C_P) * (V / ((RPM / 60) * 2 * R)); %
      Propeller efficiency
116
117 % Calculate torque coefficient C_Q
118 C_Q = C_P / (2 * pi); % Torque coefficient
119
120 % Calculate torque Q
121 Q = C_Q * (rho * (RPM / 60) ^ 2 * (2 * R) ^ 5); % Torque
122 end

```

B Parameter selection

B.1 Initial Reynolds number and Mach number distribution

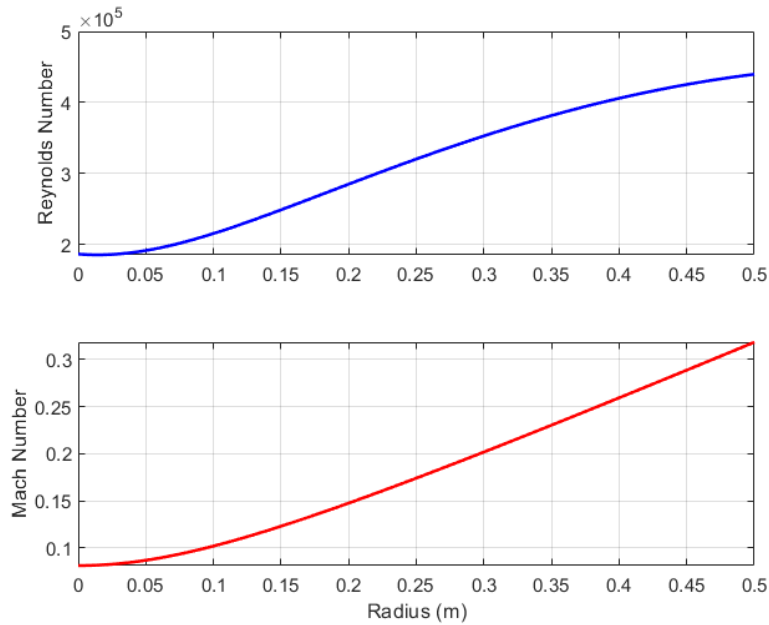


Figure 17. Initial Reynolds number and Mach number distribution along a single propeller blade

B.2 RPM selection

Note: In this plot, each point refers to a different geometry and the design point shown here is where the efficiency reaches a peak for loiter. This gives us the optimum RPM of 2000 and corresponding chord and twist distribution for this combination is selected.

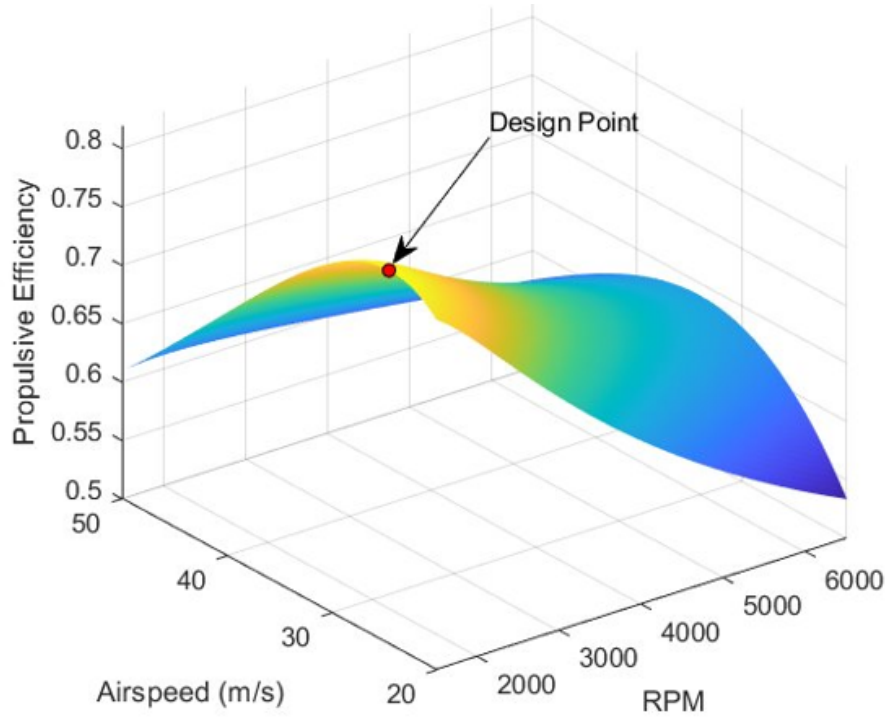


Figure 18. Efficiency against speed-RPM using optimisation code

C Aerofoil Data from airfoil tools

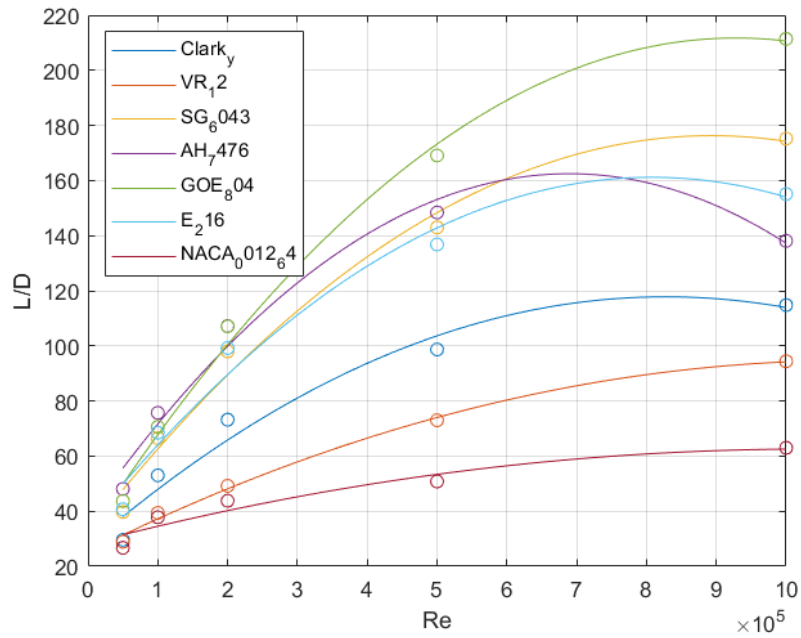


Figure 19. $\frac{L}{D}$ against Re for different aerofoils considered

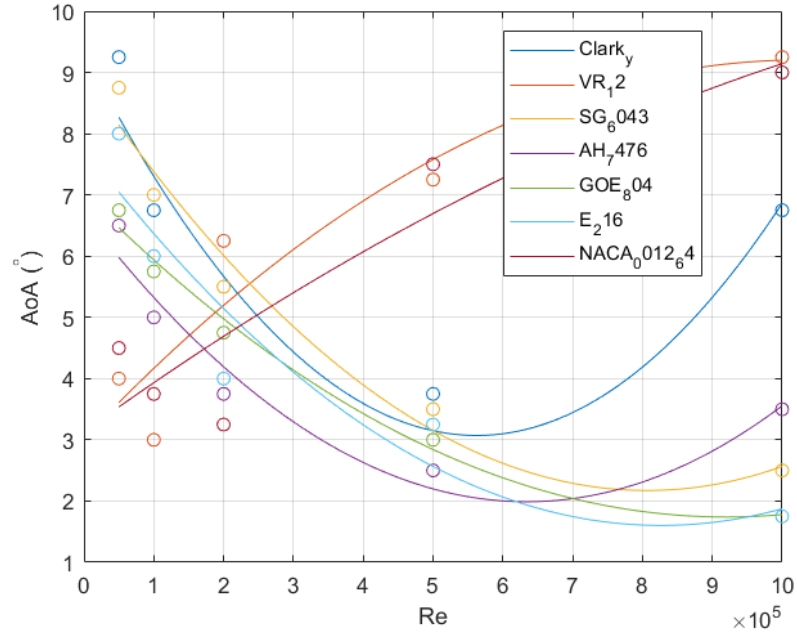


Figure 20. Angle of attack at which the max L/D is achieved against Re for different aerofoils considered

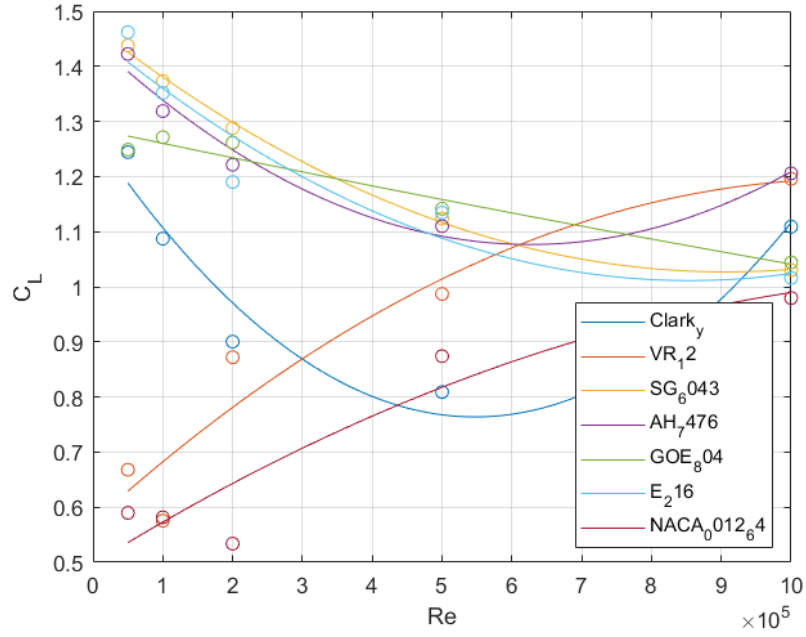


Figure 21. C_L at which the maximum L/D is achieved against Re for different aerofoils considered

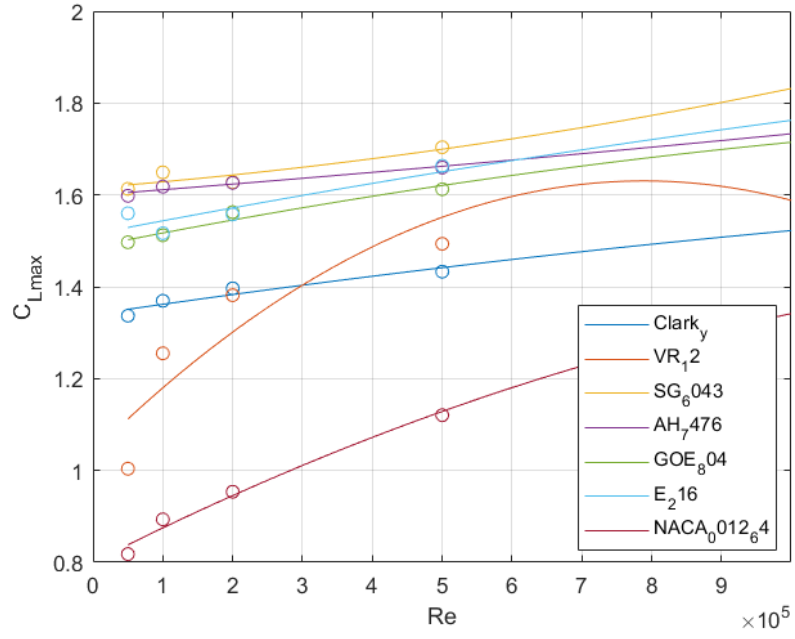


Figure 22. $C_{L_{max}}$ against Re for different aerofoils considered

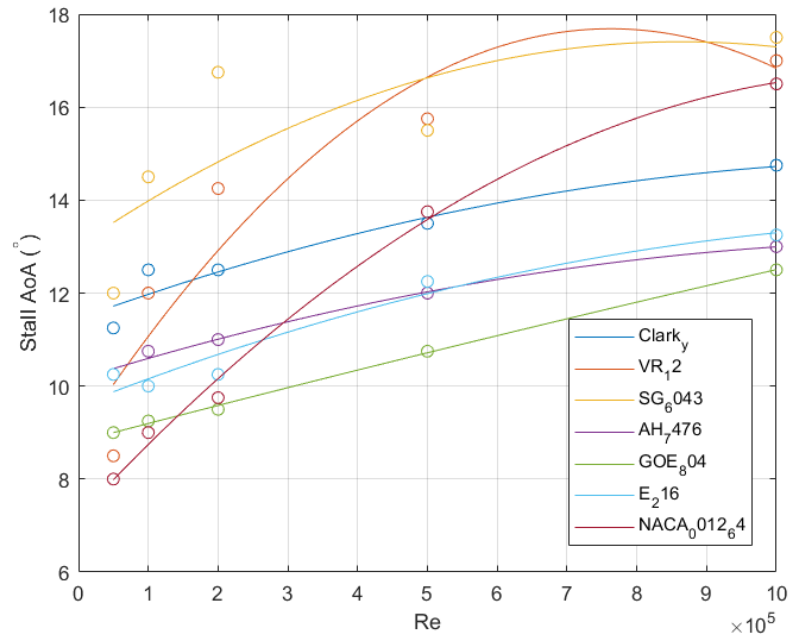


Figure 23. Stall angle at which maximum $C_{L_{max}}$ is achieved against Re for different aerofoils considered

D Final blade profile and data

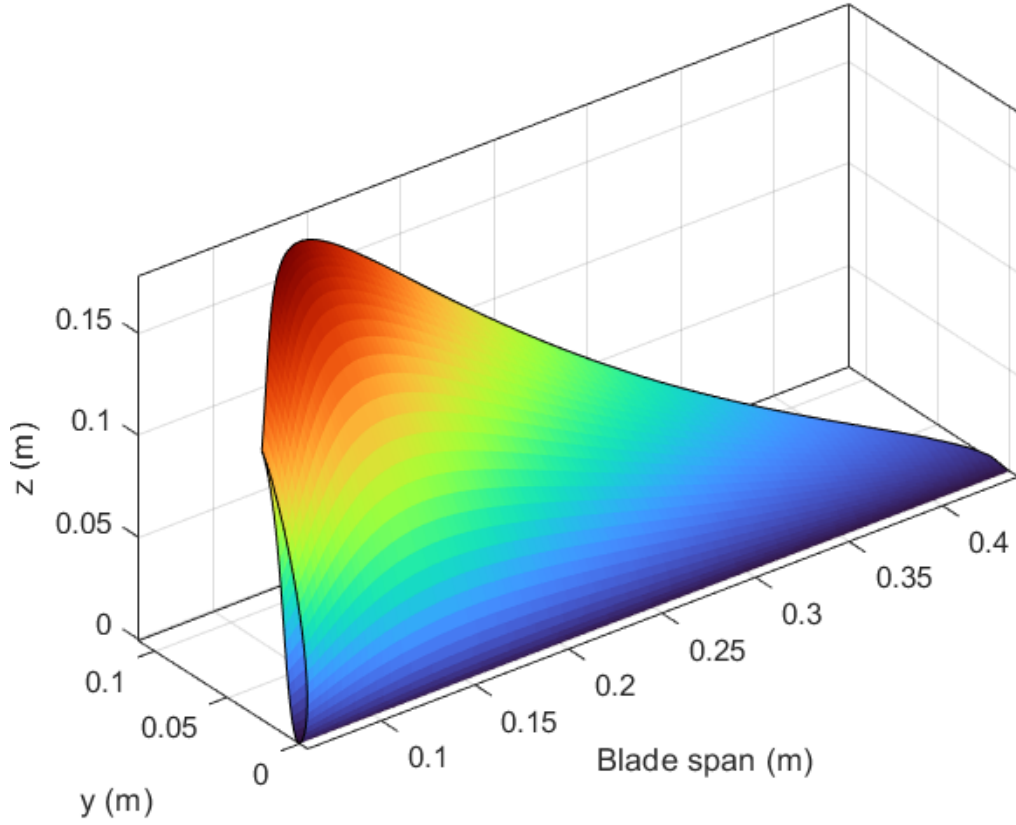


Figure 24. Finalised single blade for the propeller

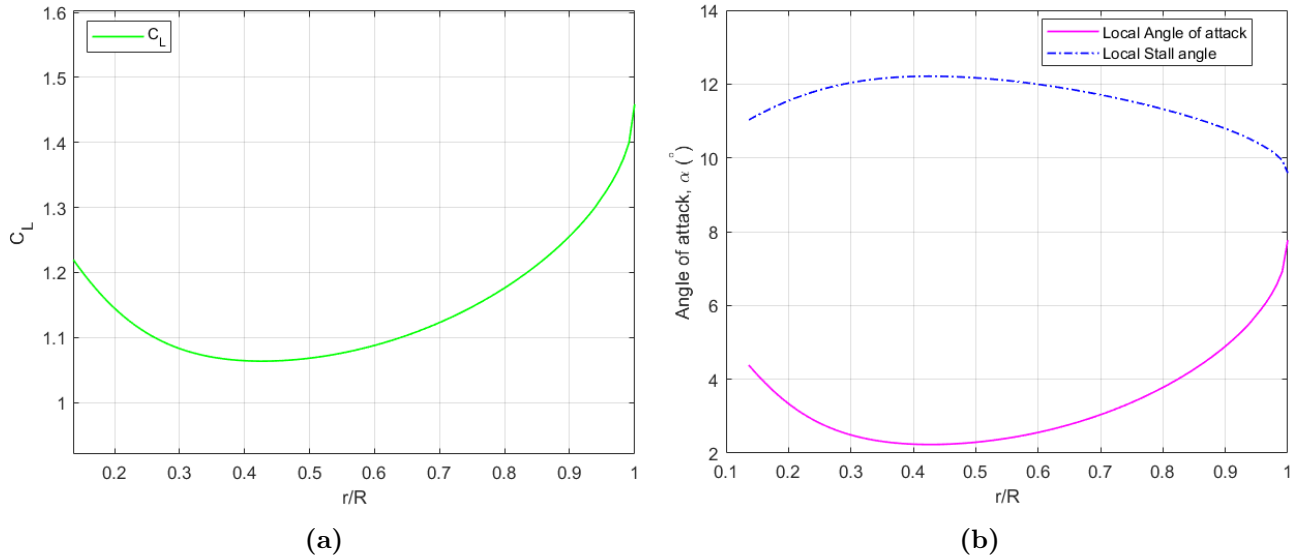


Figure 25. Optimised local C_L and local angle of attack distribution along a single propeller blade

E XROTOR results for optimised geometry

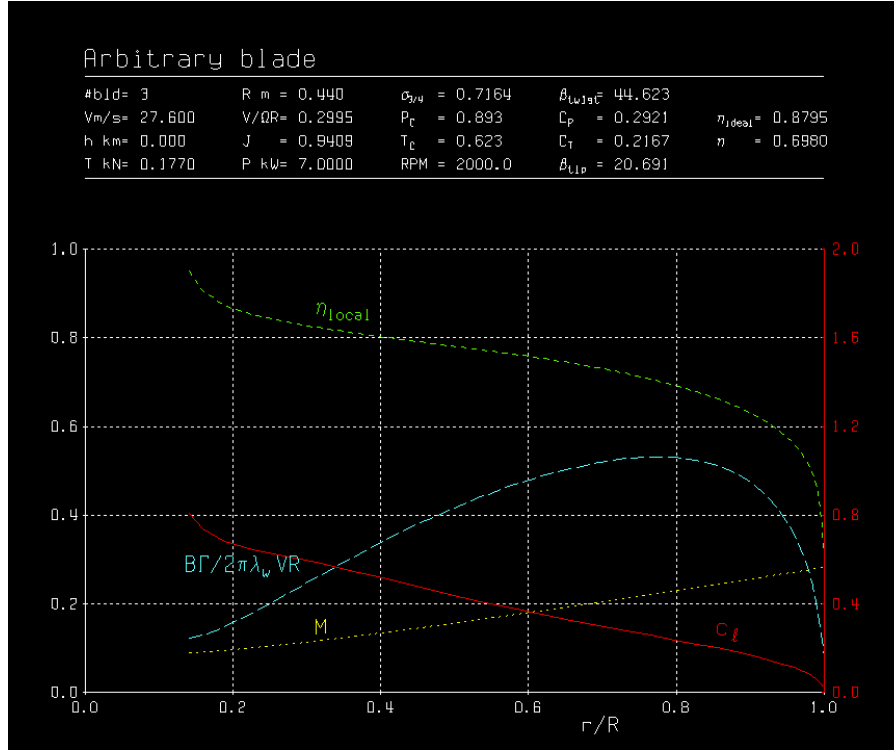


Figure 26. Finalised single blade for the propeller

F Performance analysis

Thrust coefficient:

$$C_T = \frac{T}{\rho n^2 D^4} \quad (18)$$

Power coefficient:

$$C_P = \frac{P}{\rho n^3 D^5} \quad (19)$$

Propeller efficiency:

$$\eta_P = J \frac{C_T}{C_P} \quad (20)$$

Advance ratio:

$$J = \frac{V}{nD} \quad (21)$$

G Other CFD results

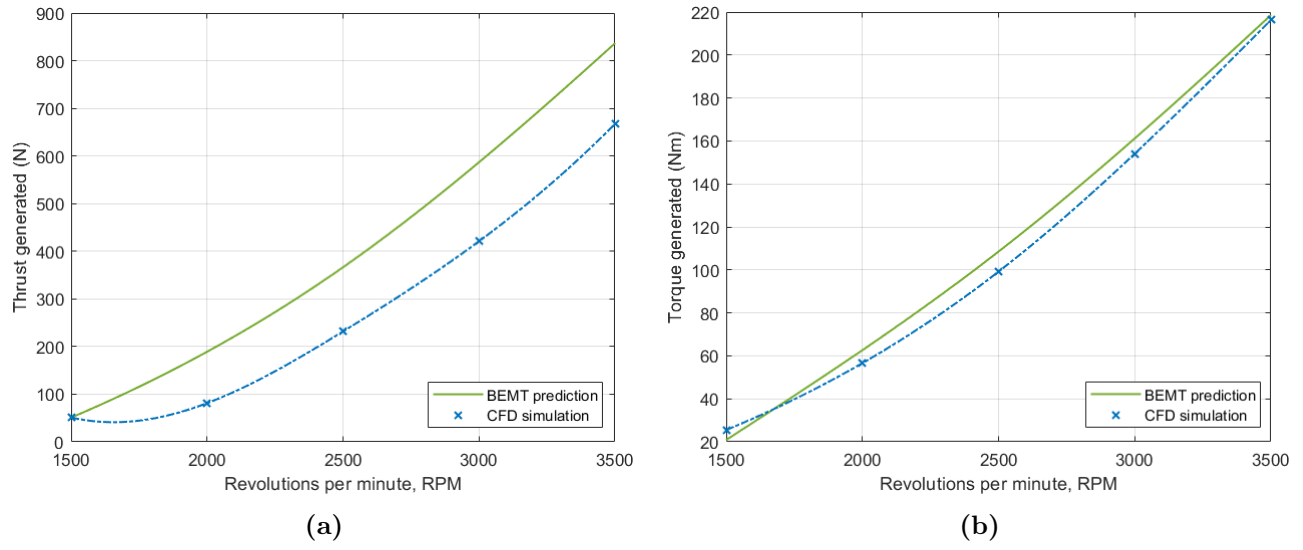


Figure 27. CFD simulation results for thrust and torque generated at different RPMs in cruise, compared to BEMT predictions